

MUHAMMAD ZULFAN ANGGADHIPUTRA

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SUMMARY

Informatics graduate from Institut Teknologi Nasional Bandung (GPA: 3.49 / 4.00) with a strong foundation in software engineering, database programming, and emerging technology R&D. Experienced in lab assistance with a proven track record of delivering IoT integrations, mobile applications, and deep learning projects.

EDUCATION

Institut Teknologi Nasional Bandung — Bandung, Jawa Barat
Bachelor of Informatics — 2022 - 2026

EXPERIENCE

Database Programming Laboratory Assistant — Sep 2024 - Jan 2025

- Served as instructor for introductory database concept and application development.
- Guided students to design, implement and intergrate databases into real-world application using Delphi7.
- Provided one-on-one mentoring to help students troubleshoot coding and database issues.

Object-Oriented Programming Laboratory Assistant — Mar 2025 - June 2025

- Served as instructor for object-oriented programming fundamentals (Java).
- Assisted students in applying OOP principles (encapsulation, inheritance, polymorphism) in software development projects.
- Mentored student teams to develop small-to-medium scale applications with best coding practices.

Computer network Laboratory Assistant — Mar 2025 - June 2025

- Served as instructor for computer networking fundamentals and configuration.
- Guided students in setting up LAN, configuring routers/switches, and implementing TCP/IP protocols.
- Supported students in network troubleshooting, security basics, and client-server project assignments.

Software Development Intern — PT Dwi Purwa Teknologi — July 2025 - Sep 2025

- Assisted in research and development (R&D) of digital solutions for client projects.
- Contributed to adding new features in an in-progress software application.
- Collaborated with senior developers to test, debug, and improve application performance.
- Gained hands-on experience in the software development lifecycle within a digital consulting environment.

IoT Programming Laboratory Assistant — Oct 2025 – Jan 2026

- Served as an instructor for IoT programming concepts, guiding students through hardware-software integration, sensors, and microcontroller implementations.
- Mentored student teams in developing IoT-based prototype projects and troubleshooting embedded systems and coding issues.
- Evaluated student assignments and lab practicals, ensuring the adherence to best development practices.

Robotics Laboratory Assistant Coordinator — Feb 2026 – June 2026

- Led and coordinated the laboratory assistant team in delivering the robotics practical curriculum, session scheduling, and lab resource management.
- Supervised practical sessions on robotics systems, sensor integration, and kinematics, providing guidance and technical troubleshooting for students.
- Oversaw student evaluations, grading standards, and final project assessments across all robotics lab sections.

PROJECTS

Remote Control Tank Sep 2024 - Jan 2025

IoT-based prototype integrating ESP32, sensors, and motor drivers with Flutter mobile control and web dashboard.

Deep Learning Image Classification with InceptionV3 Mar 2024 - June 2025

Developed and evaluated deep learning models using InceptionV3 for image classification tasks. Applied transfer learning and fine-tuning techniques on datasets.

FoodRescue: Food Waste Management Web Application Mar 2025 – July 2025

- Award: Achieved the Bronze Medal and Finalist position at the IDEA FEST 3 National Competition organized by STIMATA and Sentosa Foundation.
- Developed a full-stack Laravel web application designed to combat food waste by connecting food donors with recipients through a streamlined, role-based platform.
- Implemented a comprehensive donation lifecycle system, allowing donors to post items, recipients to claim them, and administrators to verify and manage the approval workflow.
- Engineered location-based features using Leaflet and geographic coordinates, enabling users to find nearby donations, calculate pickup distances, and interact with dynamic maps.

Automated OMR Grading Web Application Oct 2025 – Jan 2026

- Community Service Project: Developed an automated Optical Mark Recognition (OMR) web system to assist teachers in grading exams efficiently, significantly reducing manual workload.
- Engineered the backend using Python and FastAPI, integrating OpenCV (Computer Vision) algorithms to accurately detect, extract, and grade filled answers from bulk-uploaded scanned LJK (Lembar Jawaban Komputer) images and PDFs.
- Built a responsive frontend using Next.js, TypeScript, and Tailwind CSS, featuring flexible exam setups, dynamic answer key inputs, and real-time grading dashboards.
- Implemented automated scoring logic and comprehensive student performance analytics, with localized data storage and auto-generated Excel report exports.

Undergraduate Thesis: Detecting Online Gambling Spam Comments on YouTube Nov 2025 – June 2026

- Conducted in-depth research to advance expertise in Machine Learning, developing an automated classification model to detect text obfuscation in Indonesian YouTube comments.
- Engineered a hybrid feature representation by combining 1,000 TF-IDF Character Trigram features with nine numerical statistical features (e.g., emoji count, digit ratio).
- Trained and optimized a Random Forest classifier using GridSearchCV, and evaluated the model robustly using 10-Fold Stratified Cross-Validation.
- Successfully achieved an outstanding model performance of 99.56% across Accuracy, Precision, Recall, and F1-Score.
- Tech Stack: Machine Learning, Random Forest, TF-IDF, GridSearchCV, Natural Language Processing (NLP).

FoodLink – Cloud-Native Food Surplus Rescue Platform Sep 2026 – Present

- Social Impact & Enterprise Platform: Developed a geolocation-based food surplus rescue and redistribution platform, utilizing Domain Services decomposition to connect donors, beneficiaries, and volunteers, with the goal of reducing food waste and carbon emissions.
- Backend, Concurrency & Event Streaming: Engineered the backend using PHP 8.3 and Laravel 12, integrating PostgreSQL 16 (with `FOR UPDATE` row-level locking), PostGIS (GIST spatial indexing), Redis 7 (implementing Lua-based distributed mutex locks and tagged caching), and Apache Kafka (utilizing the Transactional Outbox pattern with three fan-out consumers) to prevent overselling and facilitate real-time event stream processing.
- Frontend, Interactive Maps & RESTful API: Built a responsive interface and a RESTful API (`/api/v1`) using Laravel Blade, Tailwind CSS, Alpine.js, Leaflet.js, and OpenAPI 3.0 (Swagger). Key features include `Idempotency-Key` protection, radius-based map searching, household coordinate fuzzing, and a QR/PIN-based handover verification system.
- DevOps, Security & Automated Analytics: Implemented automated ESG impact analytics, EXIF-stripped WebP image processing for stateless S3/MinIO storage, Docker containerization, and Kubernetes orchestration (featuring zero-downtime `RollingUpdate` deployments, Horizontal Pod Autoscaling, and `/healthz` probes). The entire system is managed via GitLab CI/CD, achieving a 100% automated test pass rate (32/32 tests).

AeroScan – Real-Time Military Aircraft Detection & Tactical Edge AI Platform Oct 2026 – Present

- Computer Vision & Defense Intelligence Platform: Developed an end-to-end tactical computer vision system to detect, classify, and track 15 distinct military combat aircraft (e.g., F-22, Su-57, B-2, J-20) from large-scale satellite/aerial imagery, enabling rapid situational awareness and automated threat identification.
- Data Engineering & Deep Learning Architecture: Engineered a high-throughput ETL pipeline converting 12 GB of per-image PASCAL VOC CSV bounding-box annotations into normalized YOLO format with geometric boundary validation; trained a custom YOLOv11s detector using transfer learning, PyTorch 2.6, Automatic Mixed Precision (AMP FP16), and Cosine Annealing learning rate schedules across a stratified 80/15/5 split (~8,400+ curated images).
- TensorRT Acceleration & Edge Optimization: Compiled the PyTorch model into an optimized NVIDIA TensorRT 10 FP16 engine via ONNX graph slimming and Ada Lovelace kernel auto-tuning on an RTX 4060 GPU (8GB VRAM), compressing weights to 22.1 MB and slashing forward inference latency to ~1.8 ms/frame (>500 FPS throughput) for ultra-low-latency edge deployment.
- Real-Time Stream Dashboard, Automated Logging & Error Analysis: Built an interactive Command & Control dashboard using Streamlit and OpenCV featuring real-time video/webcam ingestion, dynamic model switching (.engine vs .pt), threshold tuning, smart cooldown auto-capture, and structured CSV detection audits; validated on unseen test data achieving up to 87.9% mAP@50 (B-52) and 82.3% mAP@50 (Su-57) alongside rigorous confusion matrix error diagnosis.

ADDITIONAL INFORMATION

- **Technical Skills:**
  a. Programming & Development: Java, Python, C, JavaScript, Node.js, Flutter, Delphi 7.
  b. Database & Networking: MySQL, Huawei Firewall, MikroTik.
  c. Machine Learning & Data Science: Random Forest, Support Vector Machine (LinearSVC), Logistic Regression, TF-IDF, Natural Language Processing (NLP).
  d. Data Visualization & Testing: Power BI, Katalon Studio, Selenium.
  e. Operating Systems: Windows, Red Hat, Ubuntu.
- **Software and Tools:** Visual Studio Code, IntelliJ IDEA, NetBeans, Android Studio, Arduino IDE, eNSP, Adobe Animate, Adobe Premiere, Canva, Figma, Microsoft Word, Microsoft Excel.
- **Certifications:** Huawei Certified ICT Associate – Security (HCIA-Security V4.0).
- **Languages:** English - Intermediate , Indonesian - Native.